CUSTOMER ANALYTICS / CAPSTONE

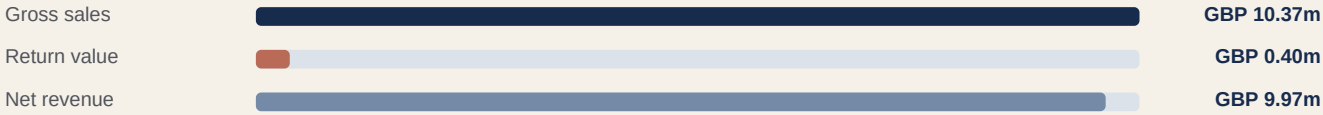
Online Retail

RFM Segmentation

From transaction-level data quality to customer groups that can guide retention, cross-sell, and reactivation experiments.

541,909	9.97m	4,336	4,322
raw transaction lines	net revenue after returns	identified customers	positive-value RFM customers

Sales retained after returns



Executive readout

The analysis establishes a traceable cleaning process, retains customer-linked returns in net value, and answers the business questions before segmentation. Rule-based RFM produces eight explainable groups. In the algorithmic comparison, K-Means with K=2 has the highest sampled silhouette (0.429) among the tested K values, but it divides customers into only two broad engagement/value groups. Use the clusters as a macro-level view and the rule-based segments for finer campaign targeting.

Method at a glance

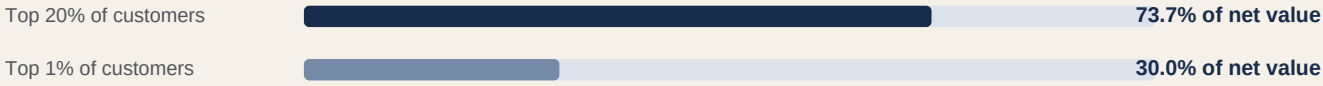
DATA QUALITY	RFM DESIGN	MODEL CHECKS
Standardized types and labels; tested missing IDs by invoice; classified duplicates, admin charges, zero-price rows, and cancellations; matched 82.5% of known-customer return lines to an earlier purchase.	Customer-level Recency, unique invoice Frequency, and net Monetary. Positive-Monetary customers are log1p transformed and standardized; CustomerID is excluded from model features.	K-Means K=2..8: inertia plus sampled silhouette. Hierarchical Ward on 1,200 customers. DBSCAN parameter checks report cluster count, silhouette excluding noise, and noise share.

Scope note: The source file records transactions from December 2010 through December 9, 2011. The findings are historical descriptions, not forecasts of present-day customer behavior.

01 / Business findings

Twenty EDA findings are recorded in the notebook. The measures below summarize the most decision-relevant patterns.

Customer value concentration



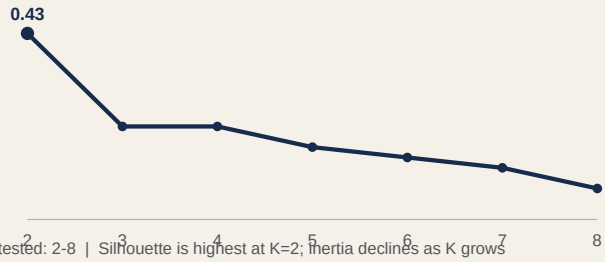
QUESTION	OBSERVED RESULT	BUSINESS IMPLICATION
Revenue and returns	Gross sales 10.37m; return value 0.40m; net revenue 9.97m. Merchandise return rate: 2.30%.	Track net value alongside gross sales; review return-heavy products and periods.
Seasonality	November 2011 was the strongest complete month for net revenue and sale orders. Sep-Nov generated 37.8% of net revenue in the 12 complete months.	Plan inventory and fulfillment before the observed peak; treat December 2011 as partial.
Customer concentration	Top 20% of known customers generated 73.7% of customer net value; top 1% generated 30.0%.	Protect high-value relationships without applying unnecessary blanket discounts.
Repeat purchase	34.4% of known customers purchased once.	Test a second-purchase journey and measure repeat conversion.
Products	21.9% of positive-revenue products generated 80% of product revenue; 23 historically strong products had no sales in the final 90 days.	Prioritize core assortment; investigate inactivity before calling items dead stock.
Geography	United Kingdom contributed 84.4% of net revenue. Netherlands was the leading non-UK market by net revenue.	Separate UK and international reporting; test international economics before expansion.
Bulk behavior	9.5% of known-customer orders met the 531-unit bulk threshold and represented 39.3% of known-order value.	Profile wholesale-like behavior separately from typical order activity.

Additional analysis in the notebook covers weekday and hourly patterns, product rankings by units and revenue, basket distributions, customer recency and frequency, guest checkout, service lines, and return patterns.

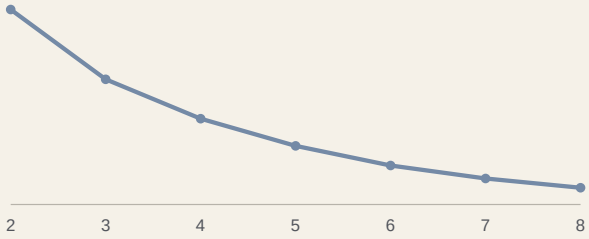
02 / Clustering and validation

Only Recency, Frequency, and Monetary are used. Each feature is transformed with log1p and standardized before distance-based modeling.

Silhouette by K



K-Means inertia (elbow)



K tested: 2-8 | Silhouette is highest at K=2; inertia declines as K grows

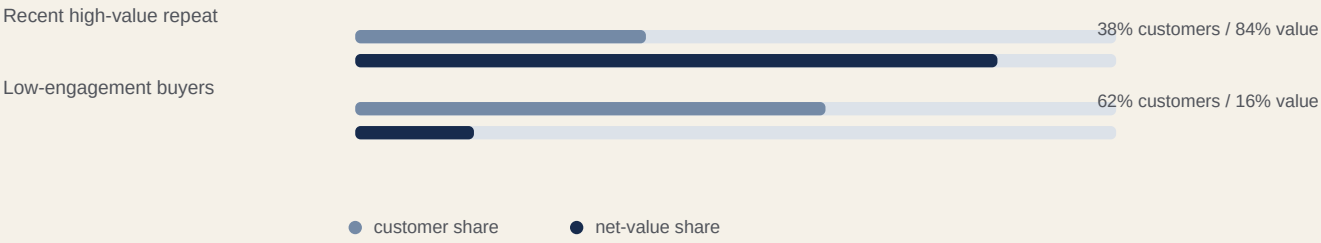
METHOD	GROUPS	SILHOUETTE	NOISE	READOUT
K-Means (selected)	2	0.429*	0%	Best sampled silhouette among K=2..8; broad two-group split.
Ward hierarchical	2	0.334*	0%	Evaluated on a reproducible 1,200-customer sample; dendrogram shown for 250.
DBSCAN	2-4	0.14-0.29*	2.36-8.81%	Density settings change cluster count; some customers are labeled noise.

* Silhouette scores are sample estimates. The K-Means evaluation used a deterministic sample of up to 2,000 customers; hierarchical clustering used 1,200; DBSCAN silhouette excludes points labeled as noise.

Selected K-Means profiles

PROFILE	CUSTOMERS	MEDIAN R / F / M	REVENUE SHARE	ACTION
Recent high-value repeat	1,652 (38.2%)	16 days / 6 / 2,024.70	84.4%	Reward loyalty and cross-sell; avoid blanket discounts.
Low-engagement buyers	2,670 (61.8%)	95 days / 1 / 355.72	15.6%	Use low-cost reactivation only when expected value supports it.

Customer share and value share



K=2 has the strongest sampled separation among tested values, but remains a broad behavioral split. Keep the rule-based RFM groups for finer campaign targeting.

03 / Recommendations and use

Protect repeat value

Pilot loyalty and relevant cross-sell treatment for recent, high-value repeat buyers. Keep discounts selective and measure net margin.

Build the second purchase

Test onboarding and product recommendations for recent, low-frequency customers; measure second-order conversion against a holdout.

Reactivate selectively

Prioritize lapsed customers using the existing At Risk rule-based RFM group and customer value. Cap offer cost and compare incremental net revenue.

Plan around the observed peak

Use the complete-month seasonal pattern to plan stock and fulfillment, while monitoring newer periods before carrying the historical pattern forward.

Protect assortment concentration

Give the high-contribution products greater availability attention. Investigate the 23 inactive-product candidates before making catalogue decisions.

Separate wholesale-like activity

Evaluate bulk-order customers with appropriate service, pricing, and fulfillment economics rather than comparing them directly with small baskets.

04 / Limitations and next steps

• The dataset is historical and ends in 2011; customer segments cannot be presumed current.

• Unidentified guest transactions are excluded from customer-level RFM. Fourteen identified customers with non-positive net Monetary are excluded from log-based clustering but retained in the wider customer analysis.

• K-Means assigns every modeled customer and favors compact, roughly spherical groups. Log scaling reduces but does not eliminate outlier effects.

• Hierarchical silhouette and dendrogram results are sampled. DBSCAN depends on `eps` and `min_samples`; noise handling affects comparisons.

• Next, test segment stability over time and across samples, validate profiles with business stakeholders, and run controlled campaigns. Monitor incremental repeat revenue, return rates, and margin before scaling actions.

Conclusion

K-Means provides a useful two-group summary, while rule-based RFM retains finer distinctions for targeted action. Neither is a churn prediction; validate both on newer data and controlled campaigns.

A responsible path from analysis to action

01 Validate

segment stability

02 Pilot

controlled campaigns

03 Scale

only on measured lift